KYRYLO POSTUPALENKO

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PERSONAL PROFILE

MSc Computer Science student at the University of St Andrews focusing on embedded systems and edge AI. Experienced in developing real-time micro-controller embedded systems written in C. Interested in solving complex challenges with resource-constrained hardware.

SKILLS

- **Embedded Systems & Robotics:** C, ESP32, FreeRTOS (Dual-core), BLE, Raspberry Pi, UART (GPS Parsing), PWM, CAD, 3D Printing
- **AI & Data Processing:** Python, NumPy, Gradio, MATLAB
- **Web:** Three.js, Node.js, React.js, Cloudflare, GitHub Actions, Docker, Linux
- **Software & Programming:** Dart, Flutter, JavaScript, Java
- **Languages:** English, Ukrainian, Russian

EDUCATION

University of St Andrews 2026 - expected 2027
MSc Computer Science
- Modules: Machine Learning, Language and Computation, Signal Processing: Sound, Image, Video

Lancaster University 2022 - 2025
BSc (Hons) Computer Science - **First-Class Honours**
- Coursework: Genetic Algorithm for the Shortest Path Problem (MATLAB), LoG Edge Detector (MATLAB)

WORK EXPERIENCE

Founder & Developer - PowerBander 2026 - Present
- Designed and prototyped the Touge Chime (an automotive mechanical speed warning device) using **CAD** design, **3D Printing** and custom fabrication.
- Architected dual-core (**FreeRTOS**), real-time embedded **C** firmware on an ESP32-Mini, splitting BLE GATT communications on Core 0 from real-time GPS parsing and hardware solenoid control on Core 1.
- Developed a garage-themed, interactive 3D e-commerce storefront utilising **Three.js,** self-hosting the full-stack infrastructure on a **Linux Raspberry Pi** server.
- Published the official iOS companion app, built with **Flutter**, providing real-time wireless control and display of sensor data of Touge Chime over **BLE** communication.
- Executed a viral organic video marketing campaign for PowerBander generating **1M+** views across platforms and **1K+** Instagram followers, capturing **300+** genuine waitlist sign-ups for Batch 1 of the Touge Chime.

Teaching Assistant - Lancaster University Jan - May 2025
- Supported undergraduate students to learn functional programming concepts by leading code debugging and providing structured feedback in laboratory sessions.
- Assisted professors in facilitating group project practical sessions and advising teams on best practices.

PROJECTS

Portfolio Website (Kyrylo.uk)
- Ubuntu/iOS simulator portfolio website with an embedded Falling Blocks arcade game and persistent high scores.
- Built using **React.js** with the backend powered by **Node.js** and **SQLite**, with **GitHub Actions** for automated deployments.

Image to STL
- Created a **Python** tool that converts images into 3D printable STL mesh models via heightmaps using **Trimesh**, **NumPy** (high-performance matrix operations) and **Pillow** with a simple **Gradio** UI.

VOLUNTEERING

Volunteer & Team Leader - Help Deliver Aid To Ukraine Mar - Jun 2022
- Managed and worked as a team of volunteers to load humanitarian aid onto lorries bound for Ukraine.
- Raised **over £7,000** through a social media campaign that promoted the fundraiser and built public awareness.

Team Member - UKRO-SAXONS 2022 - 2024
- Transitioned to sending 4x4 vehicles full of humanitarian aid after **5** lorries worth of aid was delivered.

INTERESTS

Video production (filming and editing) Classic JDM Cars Volleyball Travelling